Eric Ding

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Education

Cornell University — Ithaca, NY, United States
Ph.D., Computer Science, Direction: Systems and Networking, GPA: 4.0 — *Aug. 2024 – June 2029 Expected*

University of Michigan — Ann Arbor, MI, United States
B.S.E., Computer Science, Summa Cum Laude, GPA: 3.96, Dual Degree Program — *Aug. 2022 – Apr. 2024*

Shanghai Jiao Tong University — Shanghai, China
B.S.E., Electrical and Computer Engineering, Outstanding Graduate, GPA: 3.76, top 10% — *Sep. 2020 – Aug. 2024*

Research Experience

Graduate Student Researcher — Ithaca, New York
Cornell University — *Aug. 2024 – Present*

- Designed RDMA networking protocols and ML workload simulator for energy-efficient GPU interconnects.
- Built profiling and benchmarking pipelines for evaluating collective communication implementation.
- Implemented straggler-mitigation collective communication algorithm for distributed ML workloads.
- Developed physical layer routing algorithms for silicon photonic fabrics in rack-scale ML networks.

Machine Learning System Research Assistant — Ann Arbor, MI, United States
SymbioticLab, University of Michigan — *May 2023 – Apr. 2024*

- Developed resource management systems for federated learning.
- Contributed to open-source project *FedScale* in federated learning optimizer implementation.

Embedded System Research Assistant — Ann Arbor, MI, United States
The Fan Lab, University of Michigan — *May 2023 – Sep. 2023*

- Developed *WASP*, a wireless wearable device, capable of monitoring sweat for early disease detection.
- Designed a reliable communication protocol that operates atop I2C and Bluetooth Low Energy protocols.

Work Experience

Research Intern, Networking — Santa Clara CA, US
NVIDIA — *May 2026 – Aug. 2026*

- Designed an optical-electrical GPU NVLink network architecture for LLM inference acceleration.
- Achieved higher tokens/s/GPU and tokens/s/watt over fat tree topology at the same user interactivity.

Artificial Intelligence Internship — Shanghai, China
Bosch — *May 2024 – Aug. 2024*

- Designed and implemented a multi-agent system using LangGraph, OpenAI endpoints, and Neo4J graph database for retrieval augmented generation and domain-specific task reasoning.

Publication

- **Eric Ding**, Barry Lyu, Bhaskar Kataria, Rachee Singh, Opus: Photonic Rail-Optimized Fabric in ML Datacenters, ACM Special Interest Group on Data Communications **(SIGCOMM 2026)**
- Abhishek Vijaya Kumar, **Eric Ding**, Arjun Devraj, Darius Bunandar, Rachee Singh, Reconfigurable Torus Fabrics for Multi-tenant ML, ACM International Conference on Architectural Support for Programming Languages and Operating Systems **(ASPLOS, 2026)**
- **Eric Ding**, Chuhan Ouyang, Rachee Singh, Photonic Rails in ML Datacenters, Twenty-Fourth ACM Workshop on Hot Topics in Networks **(HotNets 2025)**
- **Eric Ding**, Rachee Singh, PipSwitch: A Circuit Switch Using Programmable Integrated Photonics, Optical Fiber Communication Conference **(OFC 2025)**
- Jiachen Liu, Fan Lai, **Eric Ding**, Yiwen Zhang, Mosharaf Chowdhury, Venn: Resource Management Across Federated Learning Jobs, The Eighth Annual Conference on Machine Learning and Systems **(MLSys 2025)**

- Anjali Devi Sivakumar, Ruchi Sharma, Chandrakalavathi Thota, **Eric Ding**, Xudong Fan,WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss, **ACS Sensors 2023**
- **Eric Ding**, Keshav Pradeep, Rachee Singh, OMRC: Multipath Reliability for Direct-Connect Optical Interconnects, Under submission
- **Eric Ding**, Byungsoo Oh, Bhaskar Kataria, Kaiwen Guo, Jelena Gvero, Abhishek Vijaya Kumar, Arjun Devraj, Lindsey Bowen, Atharv Sonwane, Emaad Manzoor, Rachee Singh, CCL-Bench 1.0: A Trace-Based Benchmark for LLM Infrastructure, Under submission
- Arjun Devraj, **Eric Ding**, Abhishek Vijaya Kumar, Robert Kleinberg, Rachee Singh, Efficient AllReduce with Stragglers, Under submission

Volunteering

- Cornell Computer Science Student-Applicants Support Program Mentor, *Oct. 2026*
- NSDI 2026 Fall AE Committee, *Jan. 2026*
- EuroSys 2026 Shadow PC, *Oct. 2025*
- SIGCOMM 2025 AE Committee, *Aug. 2025*
- MLSys 2025 AE Reviewer, *Mar. 2025*
- Lunch and Learn Czar, *Computer Science Department, Cornell University, Feb. 2025 - May. 2025*
- Computer Science Peer Tutor, *School of Information, University of Michigan, Feb. 2024 - May. 2024*

Teaching Experience

- TA, Systems for Large-scale ML (CS5470), Cornell University, Fall 2025. Responsible for assignment development: AI inference performance benchmarking, request scheduling and batching mechanisms, CUDA kernel optimization.

Projects

A Fully Synthesizable Out-of-Order N-Way RISC-V Processor | *SystemVerilog, Verdi*
CNN Convolution Optimization | *CUDA, Slurm*
Operating System Kernel (CPU Scheduler, Thread Library, Memory Management, File System) | *C++*
Distributed Search Engine (MapReduce, PageRank) | *Python, React, AWS*

Grants and Awards

- Graduate School Fellowship, Cornell University, Feb. 2024
- James B. Angell Scholar, University of Michigan, Feb. 2024
- University Honors, University of Michigan, Dec. 2023, Apr. 2023, Dec. 2022
- Dean's List, University of Michigan, Dec. 2023, Apr. 2023, Dec. 2022
- Tang Junyuan JI Scholarship Nominee, Aug. 2022
- Shanghai Jiao Tong University Merit Student Award, Nov. 2021

Technical Skills

Languages: C++, Python, C, SystemVerilog, Go, RISC-V, Matlab, Javascript, Bash, SQL, R
Frameworks: CUDA, NCCL, PyTorch, Tensorflow, JAX, vLLM, DeepSpeed, TorchTitan, LangGhain, React, gRPC, Hadoop
Developer Tools: Git, Docker, Kubernetes, Redis, Nvidia Nsight, LaTeX, Neo4J, VSCode, Arduino, STM32CubeIDE, Wireshark, Gurobi
Simulation and Modeling: Verdi, Catia, Matlab, Mathematica, LabVIEW, Pspice, Proteus, Vivado, ASTRA-sim, ns-3, NSX, gem5
Selected courses: EECS470 Computer Architecture, EECS 471 Applied Parallel Programming with GPUs, EECS482 Operating System, EECS489 Computer Network